Chapter 10. Nested and Split-Plot Designs

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1 Introduction

- Multifactor experiments with important industrial applications.
- They frequently involve one or more random factors, so the methodology of Ch. 9 (expected mean squares, variance components) is essential.
- Many variations exist; we consider the basic situations.

2 The Two-Stage Nested Design

2.1 Crossed vs nested

In a **nested** (hierarchical) design the levels of factor B are similar but *not identical* at different levels of factor A: batch 1 of supplier 1 has nothing to do with batch 1 of supplier 2.

Derivation: layout

	1	2	3	4	5	6	7	8	9	10	11	12
Supplier 1	○	○	○	○								
Supplier 2					○	○	○	○				
Supplier 3									○	○	○	○

Each supplier has its own 4 batches: 12 distinct batches, labeled 1–4 within each supplier.

Example 10.1: Purity of raw material

A company buys raw material from 3 suppliers; the material comes in batches. Is purity uniform? Select 4 batches at random from each supplier and make 3 purity determinations on each batch (coded data):

Supplier	1				2				3			
Batch	1	2	3	4	1	2	3	4	1	2	3	4
	1	−2	−2	1	1	0	−1	0	2	−2	1	3
	−1	−3	0	4	−2	4	0	3	4	0	−1	2
	0	−4	1	0	−3	2	−2	2	0	2	2	1
$y_{ij\cdot}$	0	−9	−1	5	−4	6	−3	5	6	0	2	6
$y_{i\cdot\cdot}$		−5				4				14		

2.2 Model and decomposition

Definition 10.2

$$y_{ijk} = \mu + \tau_i + \beta_{j(i)} + \varepsilon_{(ij)k}, \qquad i = 1, \ldots, a;\ j = 1, \ldots, b;\ k = 1, \ldots, n,$$

$j(i)$: the jth level of B nested under the ith level of A; each replicate is nested within the (i, j) combination, hence $\varepsilon_{(ij)k}$. There is **no** $A \times B$ interaction (every level of B appears with only one level of A).

Theorem 10.3

$$\sum_i\sum_j\sum_k (y_{ijk}-\bar y_{...})^2 = bn\sum_i(\bar y_{i..}-\bar y_{...})^2 + n\sum_i\sum_j(\bar y_{ij.}-\bar y_{i..})^2 + \sum_i\sum_j\sum_k(y_{ijk}-\bar y_{ij.})^2,$$

$$SS_T = SS_A + SS_{B(A)} + SS_E, \quad abn-1 = (a-1)+a(b-1)+ab(n-1).$$

Proof [Supplement]. $y_{ijk}-\bar y_{...} = (\bar y_{i..}-\bar y_{...})+(\bar y_{ij.}-\bar y_{i..})+(y_{ijk}-\bar y_{ij.})$. Cross products:

$$\sum_i\sum_j\sum_k(\bar y_{i..}-\bar y_{...})(\bar y_{ij.}-\bar y_{i..}) = n\sum_i(\bar y_{i..}-\bar y_{...})\underbrace{\sum_j(\bar y_{ij.}-\bar y_{i..})}_{=0} = 0,$$

and any product with $(y_{ijk}-\bar y_{ij.})$ vanishes after summing over k. DF of $B(A)$: within each of the a levels of A, $b-1$ df, total $a(b-1)$. Note $SS_{B(A)} = SS_B + SS_{AB}$ of a (wrongly) crossed analysis: $a(b-1) = (b-1)+(a-1)(b-1)$. ■

Source	SS	DF	MS
A	SS_A	$a-1$	MS_A
B within A	$SS_{B(A)}$	$a(b-1)$	$MS_{B(A)}$
Error	SS_E	$ab(n-1)$	MS_E
Total	SS_T	$abn-1$	

2.3 Expected mean squares: three cases

Case 1: A fixed, B fixed ($\sum_i\tau_i=0$, $\sum_j\beta_{j(i)}=0$ for each i).

Term	F a i	F b j	R n k	EMS
τ_i	0	b	n	$\sigma^2 + bn\sum\tau_i^2/(a-1)$
$\beta_{j(i)}$	1	0	n	$\sigma^2 + n\sum\sum\beta_{j(i)}^2/[a(b-1)]$
$\varepsilon_{(ij)k}$	1	1	1	σ^2

$H_0:\tau_i=0$: $F_0 = MS_A/MS_E$; $H_0:\beta_{j(i)}=0$: $F_0 = MS_{B(A)}/MS_E$. Estimates $\hat\tau_i = \bar y_{i..}-\bar y_{...}$, $\hat\beta_{j(i)} = \bar y_{ij.}-\bar y_{i..}$, $\hat\sigma^2 = MS_E$.

Case 2: A random, B random ($\tau_i\sim N(0,\sigma_\tau^2)$, $\beta_{j(i)}\sim N(0,\sigma_\beta^2)$).

Term	R i	R j	R k	EMS
τ_i	1	b	n	$\sigma^2 + n\sigma_\beta^2 + bn\sigma_\tau^2$
$\beta_{j(i)}$	1	1	n	$\sigma^2 + n\sigma_\beta^2$
$\varepsilon_{(ij)k}$	1	1	1	σ^2

$H_0:\sigma_\tau^2=0$: $F_0 = MS_A/MS_{B(A)}$; $H_0:\sigma_\beta^2=0$: $F_0 = MS_{B(A)}/MS_E$. $\hat\sigma^2 = MS_E$, $\hat\sigma_\beta^2 = \frac{MS_{B(A)}-MS_E}{n}$, $\hat\sigma_\tau^2 = \frac{MS_A-MS_{B(A)}}{bn}$.

Case 3: A fixed, B random (mixed).

Term	F i	R j	R k	EMS
τ_i	0	b	n	$\sigma^2 + n\sigma^2_\beta + bn\sum \tau_i^2/(a-1)$
$\beta_{j(i)}$	1	1	n	$\sigma^2 + n\sigma^2_\beta$
$\varepsilon_{(ij)k}$	1	1	1	σ^2

$H_0 : \tau_i = 0$: $F_0 = MS_A/MS_{B(A)}$; $H_0 : \sigma^2_\beta = 0$: $F_0 = MS_{B(A)}/MS_E$. $\hat\tau_i = \bar y_{i\cdot\cdot} - \bar y_{\cdots}$, $\hat\sigma^2 = MS_E$, $\hat\sigma^2_\beta = \frac{MS_{B(A)} - MS_E}{n}$.

Direct derivation of $\mathrm{E}(MS_A)$ in Case 2 [Supplement]. $\bar y_{i\cdot\cdot} = \mu + \tau_i + \bar\beta_{\cdot(i)} + \bar\varepsilon_{i\cdot\cdot}$ with $\bar\beta_{\cdot(i)} = \frac{1}{b}\sum_j \beta_{j(i)}$; the $\bar\beta_{\cdot(i)}$ are iid with variance σ^2_β/b (each supplier has its own batches). So

$$\mathrm{E}(MS_A) = bn\Big[\mathrm{Var}(\tau_i) + \mathrm{Var}(\bar\beta_{\cdot(i)}) + \mathrm{Var}(\bar\varepsilon_{i\cdot\cdot})\Big] = bn\Big[\sigma^2_\tau + \frac{\sigma^2_\beta}{b} + \frac{\sigma^2}{bn}\Big] = bn\sigma^2_\tau + n\sigma^2_\beta + \sigma^2.$$

Similarly $\bar y_{ij\cdot} - \bar y_{i\cdot\cdot} = (\beta_{j(i)} - \bar\beta_{\cdot(i)}) + (\bar\varepsilon_{ij\cdot} - \bar\varepsilon_{i\cdot\cdot})$ gives $\mathrm{E}(MS_{B(A)}) = n(\sigma^2_\beta + \sigma^2/n) = n\sigma^2_\beta + \sigma^2$. ■

Example 10.4: Purity data, line by line

$\sum y^2 = 153$, $y_{\cdots} = 13$, $y^2_{\cdots}/36 = 4.694$.

$$\begin{aligned}
SS_T &= 153 - 4.694 = 148.31,\\
SS_A &= \frac{(-5)^2 + 4^2 + 14^2}{12} - 4.694 = \frac{237}{12} - 4.694 = 19.750 - 4.694 = 15.06,\\
SS_{B(A)} &= \frac{\sum_i \sum_j y^2_{ij\cdot}}{3} - \frac{\sum_i y^2_{i\cdot\cdot}}{12}\\
&= \frac{0^2 + (-9)^2 + (-1)^2 + 5^2 + (-4)^2 + 6^2 + (-3)^2 + 5^2 + 6^2 + 0^2 + 2^2 + 6^2}{3} - \frac{237}{12}\\
&= \frac{269}{3} - 19.75 = 69.92,\\
SS_E &= 148.31 - 15.06 - 69.92 = 63.33.
\end{aligned}$$

Source	SS	DF	MS	F_0 A,B fixed	F_0 A fixed, B random	F_0 A,B random
Suppliers	15.06	2	7.53	$\frac{7.53}{2.64} = 2.85$	$\frac{7.53}{7.77} = 0.97$	0.97
Batches(Suppliers)	69.92	9	7.77	$\frac{7.77}{2.64} = 2.94$	2.94	2.94
Error	63.33	24	2.64			
Total	148.31	35				

With suppliers fixed and batches random (the natural model): $F_0 = 0.97 < F_{0.05,2,9} = 4.26$ — no supplier difference; $F_0 = 2.94 > F_{0.05,9,24} = 2.30$ — significant batch-to-batch variability within suppliers. $\hat\sigma^2_\beta = \frac{7.77 - 2.64}{3} = 1.71$, $\hat\sigma^2 = 2.64$: the batch variance is a sizeable part of total variability — the suppliers should reduce batch-to-batch variation.
Warning: using a factorial (crossed) analysis here would wrongly split $SS_{B(A)}$ into SS_B and SS_{AB}.

Example 10.5: Plants and leaves

Three plants (random), two leaves per plant (random, nested), n determinations per leaf: $y_{ijk} = \mu + \tau_i + \beta_{j(i)} + \varepsilon_{(ij)k}$, $\tau_i \sim N(0, \sigma^2_\tau)$, $\beta_{j(i)} \sim N(0, \sigma^2_\beta)$, $\varepsilon \sim N(0, \sigma^2)$; leaves are *nested* in plants (leaf 1 of plant 1 $\neq$ leaf 1 of plant 2), both random.

Source	DF	MS	EMS	F_0
Plants	2	MS_A	$\sigma^2 + n\sigma^2_\beta + 2n\sigma^2_\tau$	$MS_A/MS_{B(A)}$
Leaves(Plants)	$3(2-1)=3$	$MS_{B(A)}$	$\sigma^2 + n\sigma^2_\beta$	$MS_{B(A)}/MS_E$
Error	$6(n-1)$	MS_E	σ^2	
Total	$6n-1$			

3 The Three-Stage Nested Design

Derivation

$$y_{ijkl} = \mu + \tau_i + \beta_{j(i)} + \gamma_{k(ij)} + \varepsilon_{(ijk)l},$$

$$y_{ijkl} - \bar{y}_{....} = (\bar{y}_{i...} - \bar{y}_{....}) + (\bar{y}_{ij..} - \bar{y}_{i...}) + (\bar{y}_{ijk.} - \bar{y}_{ij..}) + (y_{ijkl} - \bar{y}_{ijk.}),$$

$SS_T = SS_A + SS_{B(A)} + SS_{C(B)} + SS_E$ with df $(a-1)$, $a(b-1)$, $ab(c-1)$, $abc(n-1)$.

EMS with A, B fixed and C random:

Term	F a i	F b j	R c k	R n l	EMS
τ_i	0	b	c	n	$\sigma^2 + n\sigma^2_\gamma + bcn\sum \tau_i^2/(a-1)$
$\beta_{j(i)}$	1	0	c	n	$\sigma^2 + n\sigma^2_\gamma + cn\sum\sum \beta^2_{j(i)}/[a(b-1)]$
$\gamma_{k(ij)}$	1	1	1	n	$\sigma^2 + n\sigma^2_\gamma$
$\varepsilon_{(ijk)l}$	1	1	1	1	σ^2

Tests: $H_0: \tau_i = 0$: $F_0 = MS_A/MS_{C(B)}$; $H_0: \beta_{j(i)} = 0$: $F_0 = MS_{B(A)}/MS_{C(B)}$; $H_0: \sigma^2_\gamma = 0$: $F_0 = MS_{C(B)}/MS_E$.

4 Designs with Both Nested and Factorial Factors

Example 10.6: Hand insertion of electronic components

Factors: 3 assembly fixtures (A, fixed), 2 workplace layouts (B, fixed), 4 operators per layout (C, random). Layouts are in different plant areas, so **operators are nested within layouts**; fixtures are crossed with both. $n = 2$.

$$y_{ijkl} = \mu + \tau_i + \beta_j + \gamma_{k(j)} + (\tau\beta)_{ij} + (\tau\gamma)_{ik(j)} + \varepsilon_{(ijk)l},$$

τ_i: fixture, β_j: layout, $\gamma_{k(j)}$: operator within layout. ($\sum \tau_i = \sum \beta_j = 0$, $\sum_i (\tau\beta)_{ij} = \sum_j (\tau\beta)_{ij} = 0$, $\gamma_{k(j)} \sim N(0, \sigma^2_\gamma)$, $(\tau\gamma)_{ik(j)} \sim N(0, \frac{a-1}{a}\sigma^2_{\tau\gamma})$ with $\sum_i (\tau\gamma)_{ik(j)} = 0$, $\varepsilon \sim N(0, \sigma^2)$.)

Term	F 3 i	F 2 j	R 4 k	R 2 l	DF	EMS
τ_i	0	2	4	2	$a-1=2$	$\sigma^2+2\sigma^2_{\tau\gamma}+16\sum\tau_i^2/2$
β_j	3	0	4	2	$b-1=1$	$\sigma^2+6\sigma^2_{\gamma}+24\sum\beta_j^2$
$\gamma_{k(j)}$	3	1	1	2	$b(c-1)=6$	$\sigma^2+6\sigma^2_{\gamma}$
$(\tau\beta)_{ij}$	0	0	4	2	$(a-1)(b-1)=2$	$\sigma^2+2\sigma^2_{\tau\gamma}+8\sum\sum(\tau\beta)_{ij}^2/2$
$(\tau\gamma)_{ik(j)}$	0	1	1	2	$(a-1)b(c-1)=12$	$\sigma^2+2\sigma^2_{\tau\gamma}$
$\varepsilon_{(ijk)l}$	1	1	1	1	$abc(n-1)=24$	σ^2

General form: $\mathrm{E}(MS_A)=\sigma^2+n\sigma^2_{\tau\gamma}+bcn\frac{\sum\tau_i^2}{a-1}$, $\mathrm{E}(MS_B)=\sigma^2+an\sigma^2_{\gamma}+acn\frac{\sum\beta_j^2}{b-1}$, $\mathrm{E}(MS_{C(B)})=\sigma^2+an\sigma^2_{\gamma}$, $\mathrm{E}(MS_{AB})=\sigma^2+n\sigma^2_{\tau\gamma}+cn\frac{\sum\sum(\tau\beta)^2}{(a-1)(b-1)}$, $\mathrm{E}(MS_{AC(B)})=\sigma^2+n\sigma^2_{\tau\gamma}$.

Source	SS	DF	MS	F_0
Fixtures	82.80	2	41.40	$41.40/5.49=7.54$
Layouts	4.08	1	4.08	$4.08/11.99=0.34$
Operators(layout)	71.91	6	11.99	$11.99/2.33=5.14$
Fixture×Layout	19.04	2	9.52	$9.52/5.49=1.73$
Fixture×Operator(layout)	65.84	12	5.49	$5.49/2.33=2.35$
Error	56.00	24	2.33	
Total	299.67	47		

Fixtures significant ($F_{0.05,2,12}=3.89$), layouts not ($F_{0.05,1,6}=5.99$), operators within layouts significant ($F_{0.05,6,24}=2.51$), fixture×operator significant ($F_{0.05,12,24}=2.18$). Variance components: $\hat\sigma^2_\gamma=\frac{11.99-2.33}{6}=1.61$, $\hat\sigma^2_{\tau\gamma}=\frac{5.49-2.33}{2}=1.58$, $\hat\sigma^2=2.33$.
Tests summary: $H_0:\tau_i=0$: $\frac{MS_A}{MS_{AC(B)}}$, $\hat\tau_i=\bar y_{i\cdots}-\bar y$; $H_0:\beta_j=0$: $\frac{MS_B}{MS_{C(B)}}$; $H_0:\sigma^2_\gamma=0$: $\frac{MS_{C(B)}}{MS_E}$, $\hat\sigma^2_\gamma=\frac{MS_{C(B)}-MS_E}{an}$; $H_0:\sigma^2_{\tau\gamma}=0$: $\frac{MS_{AC(B)}}{MS_E}$, $\hat\sigma^2_{\tau\gamma}=\frac{MS_{AC(B)}-MS_E}{n}$; $H_0:(\tau\beta)_{ij}=0$: $\frac{MS_{AB}}{MS_{AC(B)}}$, $\widehat{(\tau\beta)}_{ij}=\bar y_{ij\cdot\cdot}-\bar y_{i\cdots}-\bar y_{\cdot j\cdot\cdot}+\bar y$.

5 The Split-Plot Design

5.1 Motivation

- *Electrical components life test*: baking time (5, 10, 15 min) × oven temperature (580, 600, 620, 640). A true CRD (random temperature, one component, random time, repeat) is impractical; instead the oven is heated once and all 9 components go in (3 per time): a **restriction on randomization**.
- The **split-plot** design is a multifactor experiment in which it is not possible to completely randomize the order of the runs.
- *Paper manufacturing*: 3 pulp preparation methods × 4 cooking temperatures, 3 replicates. A CRD needs $3\times12=36$ pulp batches; pulp preparation is **hard to change**. Split-plot: per replicate and method, one batch split into 4 samples cooked at the 4 temperatures: $3\times3=9$ batches.
- Each replicate (block) is divided into three **whole plots**; pulp method is the **whole-plot treatment**.

- Each whole plot is divided into four **subplots** (split plots); temperature is the **subplot treatment**.
- Generally the hard-to-change factor is assigned to the whole plots.

Example 10.7: Paper tensile strength data

	Rep 1			Rep 2			Rep 3		
Temp	P1	P2	P3	P1	P2	P3	P1	P2	P3
200	30	34	29	28	31	31	31	35	32
225	35	41	26	32	36	30	37	40	34
250	37	38	33	40	42	32	41	39	39
275	36	42	36	41	40	40	40	44	45

5.2 Split-plot model I (full model)

Derivation

$$y_{ijk} = \mu + \underbrace{\tau_i + \beta_j + (\tau\beta)_{ij}}_{\text{whole-plot part}} + \underbrace{\gamma_k + (\tau\gamma)_{ik} + (\beta\gamma)_{jk} + (\tau\beta\gamma)_{ijk}}_{\text{subplot part}} + \varepsilon_{(ijk)1},$$

$i = 1, \ldots, r$ (blocks/replicates, random), $j = 1, \ldots, a$ (factor A, whole plot, fixed), $k = 1, \ldots, b$ (factor B, subplot, fixed), one observation per cell.

- Whole-plot part: τ_i block effect ($\sim N(0, \sigma^2_\tau)$); β_j effect of A ($\sum \beta_j = 0$); $(\tau\beta)_{ij}$ whole-plot error ($\sim N(0, \sigma^2_{\tau\beta})$).
- Subplot part: γ_k effect of B ($\sum \gamma_k = 0$); $(\tau\gamma)_{ik}$ block×B ($\sim N(0, \sigma^2_{\tau\gamma})$); $(\beta\gamma)_{jk}$ A×B ($\sum_j = \sum_k = 0$); $(\tau\beta\gamma)_{ijk}$ subplot error ($\sim N(0, \sigma^2_{\tau\beta\gamma})$).

Decomposition:

$$\begin{aligned} y_{ijk} - \bar{y}_{\cdots} = {} & (\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots}) + (\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots}) + (\bar{y}_{\cdot\cdot k} - \bar{y}_{\cdots}) + (\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots}) \\ & + (\bar{y}_{i\cdot k} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot\cdot k} + \bar{y}_{\cdots}) + (\bar{y}_{\cdot jk} - \bar{y}_{\cdot j\cdot} - \bar{y}_{\cdot\cdot k} + \bar{y}_{\cdots}) \\ & + (y_{ijk} - \bar{y}_{ij\cdot} - \bar{y}_{i\cdot k} - \bar{y}_{\cdot jk} + \bar{y}_{i\cdot\cdot} + \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdot\cdot k} - \bar{y}_{\cdots}), \end{aligned}$$

$$SS_T = SS_{\text{Block}} + SS_A + SS_B + SS_{E(w)} + SS_{\text{Block}\times B} + SS_{AB} + SS_{E(s)},$$

$$rab - 1 = (r-1) + (a-1) + (b-1) + (r-1)(a-1) + (r-1)(b-1) + (a-1)(b-1) + (r-1)(a-1)(b-1).$$

	R	F	F	R	
	r	a	b	1	
Term	i	j	k	l	EMS
τ_i	1	a	b	1	$\sigma^2 + ab\sigma^2_\tau$
β_j	r	0	b	1	$\sigma^2 + b\sigma^2_{\tau\beta} + rb\sum \beta_j^2/(a-1)$
$(\tau\beta)_{ij}$	1	0	b	1	$\sigma^2 + b\sigma^2_{\tau\beta}$
γ_k	r	a	0	1	$\sigma^2 + a\sigma^2_{\tau\gamma} + ra\sum \gamma_k^2/(b-1)$
$(\tau\gamma)_{ik}$	1	a	0	1	$\sigma^2 + a\sigma^2_{\tau\gamma}$
$(\beta\gamma)_{jk}$	r	0	0	1	$\sigma^2 + \sigma^2_{\tau\beta\gamma} + r\sum\sum(\beta\gamma)^2_{jk}/[(a-1)(b-1)]$
$(\tau\beta\gamma)_{ijk}$	1	0	0	1	$\sigma^2 + \sigma^2_{\tau\beta\gamma}$
$\varepsilon_{(ijk)l}$	1	1	1	1	σ^2 (not estimable)

Since there is one observation per cell, ε has 0 df and is inseparable from $(\tau\beta\gamma)_{ijk}$ (i.e. σ^2 is confounded with $\sigma^2_{\tau\beta\gamma}$). Tests:

- A: $MS_A/MS_{(\tau\beta)}$ (whole-plot error);
- B: $MS_B/MS_{(\tau\gamma)}$;
- AB: $MS_{AB}/MS_{(\tau\beta\gamma)}$;
- blocks: no exact test.

5.3 Split-plot model II (alternate model)

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \gamma_k + (\beta\gamma)_{jk} + \varepsilon_{ijk},$$

assuming block×B and block×A×B interactions are negligible, so they are pooled into the subplot error ε_{ijk} ($\tau_i \sim N(0,\sigma^2_\tau)$, $\sum\beta_j = 0$, $(\tau\beta)_{ij} \sim N(0,\sigma^2_{\tau\beta})$, $\sum\gamma_k = 0$, $\sum_j(\beta\gamma)_{jk} = \sum_k(\beta\gamma)_{jk} = 0$, $\varepsilon_{ijk} \sim N(0,\sigma^2)$).

$$\begin{aligned} y_{ijk} - \bar y_{\cdots} &= (\bar y_{i\cdot\cdot} - \bar y_{\cdots}) + (\bar y_{\cdot j\cdot} - \bar y_{\cdots}) + (\bar y_{ij\cdot} - \bar y_{i\cdot\cdot} - \bar y_{\cdot j\cdot} + \bar y_{\cdots}) \\ &\quad + (\bar y_{\cdot\cdot k} - \bar y_{\cdots}) + (\bar y_{\cdot jk} - \bar y_{\cdot j\cdot} - \bar y_{\cdot\cdot k} + \bar y_{\cdots}) + (y_{ijk} - \bar y_{ij\cdot} - \bar y_{\cdot jk} + \bar y_{\cdot j\cdot}), \end{aligned}$$

$$SS_T = \underbrace{SS_{\text{Block}} + SS_A + SS_{E(w)}}_{\text{whole plot}} + \underbrace{SS_B + SS_{AB} + SS_{E(s)}}_{\text{subplot}},$$

$$rab - 1 = (r-1) + (a-1) + (r-1)(a-1) + (b-1) + (a-1)(b-1) + a(r-1)(b-1).$$

(Check: $(r-1)+(a-1)+(r-1)(a-1) = ra-1$ and $(b-1)[1+(a-1)+a(r-1)] = (b-1)ar$; sum $= rab-1$.)

Term	R i	F j	F k	R l	EMS
τ_i	1	a	b	1	$\sigma^2 + ab\sigma^2_\tau$
β_j	r	0	b	1	$\sigma^2 + b\sigma^2_{\tau\beta} + rb\sum\beta_j^2/(a-1)$
$(\tau\beta)_{ij}$	1	0	b	1	$\sigma^2 + b\sigma^2_{\tau\beta}$
γ_k	r	a	0	1	$\sigma^2 + ra\sum\gamma_k^2/(b-1)$
$(\beta\gamma)_{jk}$	r	0	0	1	$\sigma^2 + r\sum\sum(\beta\gamma)_{jk}^2/[(a-1)(b-1)]$
$\varepsilon_{(ijk)}$	1	1	1	1	σ^2

Tests: A: $MS_A/MS_{E(w)}$; B: $MS_B/MS_{E(s)}$; AB: $MS_{AB}/MS_{E(s)}$.

Remark 10.8

The whole-plot factor is tested against the (usually larger) whole-plot error with few df, so it is less precise; subplot factor and interaction are more precise. Put the factor of most interest in the subplots.

Example 10.9: Paper data, model II, line by line

$y_{\cdots} = 1297$, $\sum y^2 = 47551$, $y_{\cdots}^2/36 = 46728.03$, $SS_T = 822.97$. Block totals $417, 423, 457$; pulp totals $428, 462, 407$; temperature totals $281, 311, 341, 364$; block×pulp totals $\begin{smallmatrix}138 & 155 & 124\\ 141 & 149 & 133\\ 149 & 158 & 150\end{smallmatrix}$.

$$SS_{\text{Block}} = \frac{417^2 + 423^2 + 457^2}{12} - 46728.03 = 77.56,$$

$$SS_A = \frac{428^2 + 462^2 + 407^2}{12} - 46728.03 = 128.39,$$
$$SS_{E(w)} = \frac{\sum\sum y_{ij\cdot}^2}{4} - 46728.03 - 77.56 - 128.39 = 36.28,$$
$$SS_B = \frac{281^2 + 311^2 + 341^2 + 364^2}{9} - 46728.03 = 434.08,$$
$$SS_{AB} = \frac{\sum_j \sum_k y_{\cdot jk}^2}{3} - 46728.03 - 128.39 - 434.08 = 75.17,$$
$$SS_{E(s)} = 822.97 - 77.56 - 128.39 - 36.28 - 434.08 - 75.17 = 71.50.$$

Stratum	Source	SS	DF	MS	F_0
Whole plot	Blocks	77.56	2	38.78	
	Pulp (A)	128.39	2	64.19	$64.19/9.07 = 7.08$
	Whole-plot error	36.28	4	9.07	
Subplot	Temperature (B)	434.08	3	144.69	$144.69/3.97 = 36.43$
	A×B	75.17	6	12.53	$12.53/3.97 = 3.15$
	Subplot error	71.50	18	3.97	
	Total	822.97	35		

Pulp: $P = 0.049$; temperature: $P < 10^{-7}$; interaction: $P = 0.027$ (same as `aov(y~pulp*temp+Error(block/pulp))`). Under model I the 18 subplot-error df would split into block×temp ($SS = 20.67$, 6 df) and block×pulp×temp (12 df).

Example 10.10: Paint dispersion

Four pigment mixes (A, fixed), three application methods (B, fixed: brushing, spraying, rolling), three days (blocks, random). A mix is prepared once per day and applied by all three methods: mix is the whole-plot factor, method is the subplot factor. $y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \gamma_k + (\beta\gamma)_{jk} + \varepsilon_{ijk}$ (i: day, j: mix, k: method).

Source	DF	F_0
Days (blocks)	2	–
Mix (A)	3	$MS_A/MS_{E(w)}$
Whole-plot error (days×mix)	6	
Method (B)	2	$MS_B/MS_{E(s)}$
Mix×Method	6	$MS_{AB}/MS_{E(s)}$
Subplot error	$a(r-1)(b-1) = 4 \cdot 2 \cdot 2 = 16$	
Total	35	

Example 10.11: R degrees of freedom

Two replicates (blocks), A with 4 levels (whole plot), B with 3 levels (subplot): `summary(aov(obs~A*B+Error(block/A)))`. Error: block — residual df $r - 1 = 1$; Error: block:A — A df 3, residual df $(r-1)(a-1) = 3$; Error: Within — B df 2, A:B df 6, residual df $a(r-1)(b-1) = 4 \cdot 1 \cdot 2 = 8$. Total $= 1 + 3 + 3 + 2 + 6 + 8 = 23 = 24 - 1$.

Other examples of split-plot structure:

(i) TV-tube screen color difference: on each of 5 days (whole plots, with 3 replicates), 4 operators measure in random order — day is the whole-plot factor, operator the subplot factor;

(ii) the paint dispersion study above.